Referee package (DRAFT) – Proposition A: the isotropic Gaussian-Hartree layer is the strict diagonal minimum, and the off-diagonal reduces to the additive-energy floor (class-wide H-LAYER closure)

TECT verification-first repository · claim B2-PROPA-HLAYER

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1. Purpose and scope

This referee document concerns Proposition A (the Prop-A folder of B2-PROPA-HLAYER): that the isotropic Gaussian-Hartree layer G_* is the strict comparison minimum, with the H-LAYER residuals closed class-wide on the primary shell-supported competitor class $\mathcal{A}_{\mathrm{adm}}$ ($T' \leq 13$). **Scope:** $\mathcal{A}_{\mathrm{adm}}$ (registered crystalline shell / shell-union, $T' \leq 13$) at the operating point. **Not claimed:** the full $\mathcal{C}_{\mathrm{full}}$ comfortable extension is the Reading-H package; here the margin on $\mathcal{A}_{\mathrm{adm}}$ is the comfortable $T' \leq 13$ one.

2. The result

The T-016..T-024 programme closes every ANALYTIC H-LAYER residual class-wide on $\mathcal{A}_{\mathrm{adm}}$: (D) the diagonal Hessian at G_* is block-diagonal in angular momentum with $l \geq 1$ curvature $\frac{1}{2}G_*^{-2} \geq \frac{1}{2}r_R^2 > 0$ and $l = 0$ adding $\frac{3}{2}u_{\mathrm{eff}} > 0$; (O) the off-diagonal reduces to the additive-energy floor with $R_{\mathrm{lead}}(13) = 0.650 < 1$; (S) the third-cumulant joint $= \times 1.097 > 1$ with the analytic threshold $T' < 60.4$.

3. Structure of the argument

(D) An arbitrary perturbation δG is decomposed in spherical harmonics; the rotation-invariant interaction curves only $l = 0$, so the Hessian is block-diagonal and strictly positive in every sector. (O) The condensate block is diagonal in the channel index (a sum of independent channel terms), so its operator norm is the worst single-channel ratio $R_{\mathrm{lead}} = 23.2(1 + K_{\mathrm{floor}})I$, bounded by the floor pin $K_{\mathrm{floor}} \leq T' \leq 13$. (S) MARGIN cancels in the certified joint, leaving a function of ρ_{lat} with joint $> 1 \iff T' < 60.4$.

4. Numerical reproduction

Nine scripts reproduce the class-wide second-cumulant stability, the Gershgorin row-sums, the diagonal convexity, the off-diagonal additive-energy floor and the constant certificate; each carries asserts and exits 0. Run them from the bundle.

5. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) *Attack the scope.* The result is confined to the scope stated in Sec. 1; a referee may push that boundary, which is fair — the package makes no claim outside it, and the precise scope is in the footer.

(β) *Attack the tier / over-claim.* No prose here is stronger than the registered grade (footer no-overclaim line); the status is REVIEW DRAFT, NOT operator-confirmed, and the claim-ledger tier is the arbiter.

(γ) *Attack reproduction / self-containment.* Each reproduction script carries self-test asserts and ships in the folder's bundle with its transitive dependencies (imports and runtime file reads, resolver v1.3.0); a referee runs them and diffs against `expected/`.

Quantitative sanity check (mandatory): the falsifier (footer) is concrete and pre-registered, so the claim is refutable by a single explicit construction; and the numerical values it cites are reproduced by the folder's scripts (all asserts pass).

6. Result footer

```
Result ID:               B2-PROPA-HLAYER / Prop-A referee package
                         (DRAFT)
Precise statement:       On the primary shell-supported class A_adm
                         (T’<=13), the T-016..T-024 programme closes
                         every analytic H-LAYER residual: (D)
                         block-diagonal isotropy Hessian (entropy floor
                         (1/2)r_R^2>0, breathing (3/2)u_eff>0); (O)
                         off-diagonal = additive-energy floor with
                         R_lead(13)=0.650<1; (S) third-cumulant
                         joint=x1.097>1 (analytic threshold T’<60.4).
Scope:                   A_adm (registered crystalline
                         shell/shell-union, T’<=13), operating point.
                         Comfortable margin.
Evidence grade:          class-wide analytic closure on A_adm (T6/T7);
                         EXECUTED (9 scripts PASS).
Reproduction command:    python
                         codes/vacuum/res5_020_classwide_secondcumulant_
                         stability.py ; hdiag_*.py ;
                         res1_hdiag_offdiag_floor.py
Falsifier:               an A_adm competitor with R_lead>=1 or
                         joint<=1; or a perturbation breaking the
                         block-diagonal positivity.
Tier before / after:     DRAFT referee package; A_adm closure T6/T7.
                         NOT operator-confirmed.
No-overclaim statement:  class-wide closure on A_adm (T’<=13,
                         comfortable); the full C_full extension is the
                         Reading-H package.
Next required action:    operator review + confirm -> PUBLISHED bundle.
```